

SQL

```
soham@Sohams-MacBook ~ % mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 9.6.0 Homebrew
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affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> show databases;
+-----+
| Database          |
+-----+
| first              |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+-----+
5 rows in set (0.032 sec)
mysql> create database mmcoe;
Query OK, 1 row affected (0.010 sec)
mysql> use mmcoe;
Database changed
mysql> show tables;
Empty set (0.002 sec)
mysql> create table dept (dept_no int primary key, dname varchar(50) not null,
location varchar(30) not null);
Query OK, 0 rows affected (0.010 sec)
mysql> create table emp(emp_id int primary key, ename varchar(50) not null, job
varchar(50) not null, salary double check (salary > 1000), dept_no int references
dept(dept_no));
Query OK, 0 rows affected (0.010 sec)
mysql> show tables;
+-----+
| Tables_in_mmcoe |
+-----+
| dept              |
| emp                |
+-----+
2 rows in set (0.002 sec)
```

```
mysql> insert into dept values (10, 'Sales', 'Mumbai'), (20, 'IT', 'Pune'), (30, 'Finance', 'Delhi'), (40, 'HR', 'Chennai');
```

Query OK, 4 rows affected (0.007 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> select * from dept;
```

```
+-----+-----+-----+
| dept_no | dname  | location |
+-----+-----+-----+
|      10 | Sales  | Mumbai   |
|      20 | IT     | Pune     |
|      30 | Finance | Delhi    |
|      40 | HR     | Chennai  |
+-----+-----+-----+
```

4 rows in set (0.000 sec)

```
mysql> insert into emp values (101, 'George', 'Manager', 4500, 10), (102, 'Mike', 'Clerk', 2300, 10), (201, 'Frank', 'Manager', 7200, 20), (202, 'James', 'Dev', 5500, 20), (203, 'Sean', 'Dev', 5200, 20), (204, 'Angel', 'QA', 4300, 20), (205, 'Pablo', 'QA', 3900, 20), (301, 'Steve', 'Manager', 6400, 30), (302, 'Derek', 'Assistant', 3600, 30), (501, 'Brayn', 'Executive', 8200, null);
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ')' at line 1

```
mysql> insert into emp values (101, 'George', 'Manager', 4500, 10), (102, 'Mike', 'Clerk', 2300, 10), (201, 'Frank', 'Manager', 7200, 20), (202, 'James', 'Dev', 5500, 20), (203, 'Sean', 'Dev', 5200, 20), (204, 'Angel', 'QA', 4300, 20), (205, 'Pablo', 'QA', 3900, 20), (301, 'Steve', 'Manager', 6400, 30), (302, 'Derek', 'Assistant', 3600, 30), (501, 'Brayn', 'Executive', 8200, null);
```

Query OK, 10 rows affected (0.004 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> select * from emp;
```

```
+-----+-----+-----+-----+-----+
| emp_id | ename  | job      | salary | dept_no |
+-----+-----+-----+-----+-----+
|      101 | George | Manager  |    4500 |      10 |
|      102 | Mike   | Clerk    |    2300 |      10 |
|      201 | Frank  | Manager  |    7200 |      20 |
|      202 | James  | Dev      |    5500 |      20 |
|      203 | Sean   | Dev      |    5200 |      20 |
|      204 | Angel  | QA       |    4300 |      20 |
|      205 | Pablo  | QA       |    3900 |      20 |
|      301 | Steve  | Manager  |    6400 |      30 |
|      302 | Derek  | Assistant |    3600 |      30 |
|      501 | Brayn  | Executive |    8200 |     NULL |
+-----+-----+-----+-----+-----+
```

10 rows in set (0.000 sec)

```
mysql> -- show all employees in dept no 20
```

```
Query OK, 0 rows affected (0.000 sec)
```

```
mysql> select * from emp where dept_no = 20;
```

emp_id	ename	job	salary	dept_no
201	Frank	Manager	7200	20
202	James	Dev	5500	20
203	Sean	Dev	5200	20
204	Angel	QA	4300	20
205	Pablo	QA	3900	20

```
5 rows in set (0.000 sec)
```

```
mysql> -- show all employees earning more than 5000
```

```
Query OK, 0 rows affected (0.000 sec)
```

```
mysql> select * from emp where salary > 5000;
```

emp_id	ename	job	salary	dept_no
201	Frank	Manager	7200	20
202	James	Dev	5500	20
203	Sean	Dev	5200	20
301	Steve	Manager	6400	30
501	Brayn	Executive	8200	NULL

```
5 rows in set (0.000 sec)
```

```
mysql> -- show all employees that do not belong to a department
```

```
Query OK, 0 rows affected (0.000 sec)
```

```
mysql> select * from emp where dept_no = null;
```

```
Empty set (0.000 sec)
```

```
mysql> select * from emp where dept_no is null;
```

emp_id	ename	job	salary	dept_no
501	Brayn	Executive	8200	NULL

```
1 row in set (0.001 sec)
```

```
mysql> -- show all employees working as manager
```

```
Query OK, 0 rows affected (0.000 sec)
```

```
mysql> select * from emp where job = 'Manager';
```

emp_id	ename	job	salary	dept_no
101	George	Manager	4500	10
201	Frank	Manager	7200	20

301	Steve	Manager	6400	30
-----	-------	---------	------	----

3 rows in set (0.001 sec)

mysql> -- show all employees working as Dev and QA

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where job = 'Dev' OR job = 'QA';

emp_id	ename	job	salary	dept_no
202	James	Dev	5500	20
203	Sean	Dev	5200	20
204	Angel	QA	4300	20
205	Pablo	QA	3900	20

4 rows in set (0.001 sec)

mysql> -- show all employees earning between 3000 and 6000

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where salary >= 3000 and salary <= 6000;

emp_id	ename	job	salary	dept_no
101	George	Manager	4500	10
202	James	Dev	5500	20
203	Sean	Dev	5200	20
204	Angel	QA	4300	20
205	Pablo	QA	3900	20
302	Derek	Assistant	3600	30

6 rows in set (0.000 sec)

mysql> -- show all employees working in either department no 10 or 20

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where dept_no = 10 OR dept_no = 20;

emp_id	ename	job	salary	dept_no
101	George	Manager	4500	10
102	Mike	Clerk	2300	10
201	Frank	Manager	7200	20
202	James	Dev	5500	20
203	Sean	Dev	5200	20
204	Angel	QA	4300	20
205	Pablo	QA	3900	20

7 rows in set (0.001 sec)

mysql> -- show all employees not working in dept 20 and 40

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where dept_no not in (20, 40);

emp_id	ename	job	salary	dept_no
101	George	Manager	4500	10
102	Mike	Clerk	2300	10
301	Steve	Manager	6400	30
302	Derek	Assistant	3600	30

4 rows in set (0.001 sec)

mysql> -- show employee name earning maximum salary

Query OK, 0 rows affected (0.000 sec)

mysql> select ename from emp where salary = MAX(salary);

ERROR 1111 (HY000): Invalid use of group function

mysql> select ename from emp where salary = (select MAX(salary) from emp);

ename
Brayn

1 row in set (0.001 sec)

mysql> -- show employee name earning minimum salary

Query OK, 0 rows affected (0.000 sec)

mysql> select ename from emp where salary = (select MIN(salary) from emp);

ename
Mike

1 row in set (0.001 sec)

mysql> -- show all employees earning more than average salary

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where salary > (select AVG(salary) from emp);

emp_id	ename	job	salary	dept_no
201	Frank	Manager	7200	20
202	James	Dev	5500	20
203	Sean	Dev	5200	20
301	Steve	Manager	6400	30
501	Brayn	Executive	8200	NULL

5 rows in set (0.000 sec)

mysql> -- show all employees earning less than average salary

Query OK, 0 rows affected (0.000 sec)

```
mysql> select * from emp where salary < (select AVG(salary) from emp);
```

emp_id	ename	job	salary	dept_no
101	George	Manager	4500	10
102	Mike	Clerk	2300	10
204	Angel	QA	4300	20
205	Pablo	QA	3900	20
302	Derek	Assistant	3600	30

5 rows in set (0.001 sec)

```
mysql> -- show all employees working in pune or delhi
```

Query OK, 0 rows affected (0.000 sec)

```
mysql> select e.* from emp e join dept d on e.dept_no = d.dept_no where d.location  
in (Pune, Delhi);
```

ERROR 1054 (42S22): Unknown column 'Pune' in 'where clause'

```
mysql> select e.* from emp e join dept d on e.dept_no = d.dept_no where d.location  
in ('Pune','Delhi');
```

emp_id	ename	job	salary	dept_no
201	Frank	Manager	7200	20
202	James	Dev	5500	20
203	Sean	Dev	5200	20
204	Angel	QA	4300	20
205	Pablo	QA	3900	20
301	Steve	Manager	6400	30
302	Derek	Assistant	3600	30

7 rows in set (0.001 sec)

```
mysql> select e.* from emp e join dept d on e.dept_no = d.dept_no where d.dname =  
"Sales";
```

emp_id	ename	job	salary	dept_no
101	George	Manager	4500	10
102	Mike	Clerk	2300	10

2 rows in set (0.001 sec)

```
mysql> -- show dept no and count of employees in each dept
```

Query OK, 0 rows affected (0.000 sec)

```
mysql> select dept_no, COUNT(*) from emp group by dept_no;
```

dept_no	COUNT(*)
---------	----------

```
+-----+-----+
|  NULL |      1 |
|    10 |      2 |
|    20 |      5 |
|    30 |      2 |
+-----+-----+
```

4 rows in set (0.001 sec)

mysql> -- show dept no and min, max, avg salary of employees in each department

Query OK, 0 rows affected (0.000 sec)

mysql> select dept_no, MIN(salary) as Minimum, MAX(salary) as Maximum, AVG(salary) as Average from emp group by dept_no;

```
+-----+-----+-----+-----+
| dept_no | Minimum | Maximum | Average |
+-----+-----+-----+-----+
|  NULL |    8200 |    8200 |    8200 |
|    10 |    2300 |    4500 |    3400 |
|    20 |    3900 |    7200 |    5220 |
|    30 |    3600 |    6400 |    5000 |
+-----+-----+-----+-----+
```

4 rows in set (0.001 sec)

mysql> -- show employee name starting with 'S'

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where ename like 'S%';

```
+-----+-----+-----+-----+-----+
| emp_id | ename | job      | salary | dept_no |
+-----+-----+-----+-----+-----+
|    203 | Sean  | Dev      | 5200   | 20      |
|    301 | Steve | Manager  | 6400   | 30      |
+-----+-----+-----+-----+-----+
```

2 rows in set (0.003 sec)

mysql> -- show employee name containing 'a'

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where ename like '%a%';

```
+-----+-----+-----+-----+-----+
| emp_id | ename | job      | salary | dept_no |
+-----+-----+-----+-----+-----+
|    201 | Frank | Manager  | 7200   | 20      |
|    202 | James | Dev      | 5500   | 20      |
|    203 | Sean  | Dev      | 5200   | 20      |
|    204 | Angel | QA       | 4300   | 20      |
|    205 | Pablo | QA       | 3900   | 20      |
|    501 | Brayn | Executive | 8200   | NULL    |
+-----+-----+-----+-----+-----+
```

6 rows in set (0.001 sec)

mysql> -- show employee name ending with k

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where ename like '%k';

emp_id	ename	job	salary	dept_no
201	Frank	Manager	7200	20
302	Derek	Assistant	3600	30

2 rows in set (0.001 sec)

mysql> -- show employee name having second char 'e'

Query OK, 0 rows affected (0.000 sec)

mysql> select * from emp where ename like '_e%';

emp_id	ename	job	salary	dept_no
101	George	Manager	4500	10
203	Sean	Dev	5200	20
302	Derek	Assistant	3600	30

3 rows in set (0.001 sec)

mysql>

mysql> -- show all employee names in descending order

Query OK, 0 rows affected (0.000 sec)

mysql> select ename from emp order by ename;

ename
Angel
Brayn
Derek
Frank
George
James
Mike
Pablo
Sean
Steve

10 rows in set (0.001 sec)

mysql> select ename from emp order by ename desc;

ename
Steve

```
| Sean |
| Pablo |
| Mike |
| James |
| George |
| Frank |
| Derek |
| Brayn |
| Angel |
```

```
+-----+
```

10 rows in set (0.001 sec)

mysql> -- show emp_no, ename, dept_no and dname for all employees

Query OK, 0 rows affected (0.000 sec)

mysql> select e.emp_id, e.ename, e.dept_no, d.dname from emp e left join dept d on e.dept_no = d.dept_no;

```
+-----+-----+-----+-----+
```

emp_id	ename	dept_no	dname
101	George	10	Sales
102	Mike	10	Sales
201	Frank	20	IT
202	James	20	IT
203	Sean	20	IT
204	Angel	20	IT
205	Pablo	20	IT
301	Steve	30	Finance
302	Derek	30	Finance
501	Brayn	NULL	NULL

```
+-----+-----+-----+-----+
```

10 rows in set (0.001 sec)

mysql> -- show emp_id, job and city of employees working in dept 20

Query OK, 0 rows affected (0.000 sec)

mysql> select e.emp_id,e.job, d.location from emp e left join dept d on e.dept_no = d.dept_no where e.dept_no = 20;

```
+-----+-----+-----+
```

emp_id	job	location
201	Manager	Pune
202	Dev	Pune
203	Dev	Pune
204	QA	Pune
205	QA	Pune

```
+-----+-----+-----+
```

5 rows in set (0.001 sec)

mysql> -- show employee not working in either of departments

```

Query OK, 0 rows affected (0.000 sec)
mysql> select * from emp where dept_no is null;
+-----+-----+-----+-----+-----+
| emp_id | ename | job      | salary | dept_no |
+-----+-----+-----+-----+-----+
| 501    | Brayn | Executive | 8200   | NULL    |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
mysql> -- show ename and dname of all emp irrespective of dept
Query OK, 0 rows affected (0.000 sec)
mysql> select e.ename, d.dname from emp e left join dept d on e.dept_no =
d.dept_no UNION select e.ename, d.dname from emp e right join dept d on e.dept_no
= d.dept_no;
+-----+-----+
| ename | dname |
+-----+-----+
| George | Sales |
| Mike   | Sales |
| Frank  | IT     |
| James  | IT     |
| Sean   | IT     |
| Angel  | IT     |
| Pablo  | IT     |
| Steve  | Finance |
| Derek  | Finance |
| Brayn  | NULL   |
| NULL   | HR     |
+-----+-----+
11 rows in set (0.003 sec)
mysql> -- setting auto commit off
Query OK, 0 rows affected (0.000 sec)
mysql> set autocommit=off;
Query OK, 0 rows affected (0.002 sec)
mysql> create table person (id int primary key, name varchar(50) not null, age int
check (age >= 18), city varchar(50) not null);
Query OK, 0 rows affected (0.008 sec)
mysql> insert into person values (1, "Sam", 23, "Melbourne"), (2, "Hammo", 27,
"Sydney"), (3, "Abby", 23, "Perth"), (4, "Akila", 24, "Colambo"), (5, "Alan", 24,
"Beijing");
Query OK, 5 rows affected (0.001 sec)
Records: 5 Duplicates: 0 Warnings: 0
mysql> savepoint A;
Query OK, 0 rows affected (0.000 sec)
mysql> insert into person values (6, "Steve", 23, "Melbourne");
Query OK, 1 row affected (0.000 sec)

```

```

mysql> savepoint B;
Query OK, 0 rows affected (0.000 sec)
mysql> insert into person values (7, "Micheal", 33, "Melbourne");
Query OK, 1 row affected (0.000 sec)
mysql> select * from person;
+----+-----+-----+-----+
| id | name  | age  | city    |
+----+-----+-----+-----+
| 1  | Sam   | 23   | Melbourne |
| 2  | Hammo | 27   | Sydney    |
| 3  | Abby  | 23   | Perth     |
| 4  | Akila | 24   | Colambo   |
| 5  | Alan  | 24   | Beijing   |
| 6  | Steve | 23   | Melbourne |
| 7  | Micheal | 33  | Melbourne |
+----+-----+-----+-----+
7 rows in set (0.000 sec)
mysql> rollback to B;
Query OK, 0 rows affected (0.000 sec)
mysql> select * from person;
+----+-----+-----+-----+
| id | name  | age  | city    |
+----+-----+-----+-----+
| 1  | Sam   | 23   | Melbourne |
| 2  | Hammo | 27   | Sydney    |
| 3  | Abby  | 23   | Perth     |
| 4  | Akila | 24   | Colambo   |
| 5  | Alan  | 24   | Beijing   |
| 6  | Steve | 23   | Melbourne |
+----+-----+-----+-----+
6 rows in set (0.000 sec)
mysql> rollback to A;
Query OK, 0 rows affected (0.000 sec)
mysql> select * from person;
+----+-----+-----+-----+
| id | name  | age  | city    |
+----+-----+-----+-----+
| 1  | Sam   | 23   | Melbourne |
| 2  | Hammo | 27   | Sydney    |
| 3  | Abby  | 23   | Perth     |
| 4  | Akila | 24   | Colambo   |
| 5  | Alan  | 24   | Beijing   |
+----+-----+-----+-----+
5 rows in set (0.000 sec)
mysql>

```

